CLINICAL STUDY PROTOCOL

A Phase 2, Open-Label, Randomized Study of Pembrolizumab in Combination with Nab-paclitaxel Versus Nab-paclitaxel Alone in Patients with Previously Untreated HER2-Negative Metastatic Breast Cancer

Protocol Number: SAMPLE-PEMBRO-001

NCT Number: NCT99999999

Sponsor: Example Oncology Research Institute

Phase: Phase 2

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Investigational Products: pembrolizumab, nab-paclitaxel

1. PROTOCOL SYNOPSIS

Title: Pembrolizumab plus nab-paclitaxel versus nab-paclitaxel alone in first-line HER2-negative metastatic breast cancer.

Study Design: This is a Phase 2, multicenter, randomized, open-label study. Eligible patients will be randomized 1:1 to Arm A (pembrolizumab plus nab-paclitaxel) or Arm B (nab-paclitaxel alone). Randomization will be stratified by PD-L1 status and hormone receptor status.

Study Population: Adult women and men with locally confirmed, previously untreated, HER2-negative metastatic breast cancer, including triple-negative breast cancer (TNBC) and hormone receptor-positive, HER2-negative disease.

Treatment Setting: First-line metastatic setting.

Sample Size: Approximately 120 patients will be enrolled.

2. STUDY OBJECTIVES AND ENDPOINTS

2.1 Primary Objective

To compare progression-free survival (PFS) between Arm A and Arm B according to RECIST v1.1 as assessed by the investigator.

2.2 Primary Endpoint

The primary endpoint is progression-free survival (PFS) per RECIST 1.1.

2.3 Secondary Endpoints

Secondary endpoints are overall survival (OS), objective response rate (ORR), duration of response (DOR), clinical benefit rate (CBR), and safety and tolerability assessed by CTCAE v5.0.

2.4 Exploratory Endpoints

Exploratory endpoints include the association of PD-L1 expression and tumor infiltrating lymphocytes (TILs) with PFS, and circulating tumor DNA (ctDNA) dynamics.

3. STUDY DESIGN

Patients will be screened within 28 days before randomization. Treatment will continue until disease progression, unacceptable toxicity, withdrawal of consent, or completion of 35 cycles of pembrolizumab. Each treatment cycle is 21 days.

4. TREATMENT ARMS AND DOSING

4.1 Arm A: Pembrolizumab plus Nab-paclitaxel

Patients in Arm A will receive pembrolizumab 200 mg administered intravenously on Day 1 of each 21-day cycle, in combination with nab-paclitaxel 100 mg/m2 administered intravenously on Days 1, 8 and 15 of each 21-day cycle.

4.2 Arm B: Nab-paclitaxel Alone

Patients in Arm B will receive nab-paclitaxel 100 mg/m2 administered intravenously on Days 1, 8 and 15 of each 21-day cycle.

4.3 Dose Modifications

Dose interruption of pembrolizumab is required for Grade 3 immune-related adverse events. Treatment must be permanently discontinued for any Grade 4 pneumonitis. Nab-paclitaxel dose will be reduced to 75 mg/m2 for Grade 3 peripheral neuropathy.

5. SELECTION OF STUDY POPULATION

5.1 Inclusion Criteria

1. Female or male participants aged 18 years or older.

2. Histologically or cytologically confirmed metastatic breast cancer that is HER2-negative (IHC 0, 1+, or IHC 2+/ISH-negative).

3. No prior systemic chemotherapy for metastatic disease. Prior neoadjuvant or adjuvant chemotherapy is allowed if completed at least 12 months before randomization.

4. Measurable disease according to RECIST v1.1.

5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.

6. Adequate organ function, including absolute neutrophil count of at least 1,500/mcL and hemoglobin of at least 9 g/dL.

7. Tumor tissue available for PD-L1 testing.

5.2 Exclusion Criteria

1. Prior treatment with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent.

2. Known active central nervous system metastases or carcinomatous meningitis.

3. Active autoimmune disease requiring systemic treatment within the past 2 years.

4. History of non-infectious pneumonitis that required steroids.

5. Known HER2-positive breast cancer or prior treatment with trastuzumab.

6. Pregnant or breastfeeding women.

7. Known history of Hepatitis B or active Hepatitis C infection.

6. SCHEDULE OF ASSESSMENTS

Tumor imaging by CT or MRI will be performed every 9 weeks. Laboratory tests including complete blood count and liver function tests will be done on Day 1 of every cycle. Electrocardiogram (ECG) will be performed at screening and as clinically indicated. Pregnancy test is required at screening for women of childbearing potential.

7. SAFETY MONITORING AND ADVERSE EVENTS

Adverse events will be graded using NCI CTCAE v5.0. Common adverse events reported with nab-paclitaxel include peripheral neuropathy, neutropenia, fatigue, and alopecia. Immune-related adverse events reported with pembrolizumab include pneumonitis, colitis, hepatitis, hypothyroidism, and rash. An independent data monitoring committee will review safety data every 6 months.

8. STATISTICAL ANALYSIS

Progression-free survival will be compared using a stratified log-rank test. Hazard ratios and 95% confidence intervals will be estimated using a stratified Cox proportional hazards model. Kaplan-Meier methodology will be used to estimate median PFS. The primary analysis will occur after approximately 90 PFS events.

9. FOLLOW-UP

After treatment discontinuation, patients will be followed for survival every 12 weeks until death, withdrawal of consent, or study end.

10. BIOMARKER ANALYSIS

PD-L1 expression will be assessed by immunohistochemistry using the combined positive score (CPS). Estrogen receptor (ER), progesterone receptor (PR) and Ki-67 will be assessed centrally.